Minimal Recurrence Orders and Integral Solution Lattices for Franel Sequences

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Abstract

The generalized Franel sequence is $A_s(n) = \sum_{k=0}^{n} \binom{n}{k}^s$. We develop two complementary rigidity arguments. The analytic argument represents $A_s(n)$ as a moment of a real Laurent polynomial on a torus. We identify every nonzero critical point, verify that its Hessian is definite, and prove that each distinct nonzero critical value gives a genuine singularity of the continued moment generating function. A recurrence of shift order r allows at most r such singularities. Together with Stoll's existence theorem this yields the order $\lfloor (s+1)/2 \rfloor$. The arithmetic argument concerns the classical cubic Franel recurrence. Its discrete Abel identity forces the Casoratian of two integral solutions to vanish, giving the complete lattice $\mathbb{Z}A_3$. The rational solutions with a common denominator are exactly $\mathbb{Q}A_3$. We also give a stronger local statement: any other rational solution has unbounded denominators at every odd prime. The two arguments are independent, and the paper records their hypotheses and provenance separately. These are research-draft proofs, not a claim of completed external peer review or independently established priority.

Keywords. Franel numbers; polynomial recurrences; Cauchy transforms; Morse critical points; Casoratian; integral solutions.

1 Two rigidity questions

For an integer $s \geq 1$, put

$$A_s(n) = \sum_{k=0}^{n} \binom{n}{k}^s, \qquad n \geq 0. \tag{1}$$

The parameter s specifies a sequence; the variable n is its index. A linear recurrence in this paper always means an identity in the variable n. We use *shift order*, rather than differential order or the number of nonzero terms. These quantities must not be interchanged.

Definition 1.1. A nonzero recurrence of shift order r for a sequence a is an identity

$$\sum_{j=0}^{r} p_j(n) a_{n+j} = 0 \tag{2}$$

for every sufficiently large integer n, where $p_j \in \mathbb{Q}(n)$ and $p_0 p_r \neq 0$. Its shift order is the difference between the largest and smallest shifts after zero endpoint coefficients have been removed. The minimum such order is denoted by $\operatorname{ord}_{\mathbb{Q}(n)} a$.

Stoll's theorem gives a recurrence for (1) of shift order at most $\lfloor (s+1)/2 \rfloor$ [1]. The distinction between this upper bound and minimality is explicit in Straub–Zudilin [2]. Their

lower bound is proved for telescoping recurrences based on the binomial-power summand. The unrestricted lower bound is a separate question. The argument below concerns an arbitrary recurrence (2); no telescoping certificate is assumed.

There is another question even when a specific recurrence is fixed. Write $S_n = A_3(n)$. The classical Franel equation is

$$(n+1)^2 f_{n+1} = (7n^2+7n+2)f_n + 8n^2 f_{n-1}, \qquad n \geq 1. \tag{3}$$

Its rational solution space has dimension two, because f_0, f_1 are free. Its integral solution set need not have rank two. Tonien's conclusion [3] asks whether the space contains integral sequences other than multiples of S. We answer that question without changing the recurrence or its starting index.

Theorem 1.2 (Minimal order; [source T39–40])**.** *For every integer $s \geq 1$,*

$$\operatorname{ord}_{\mathbb{Q}(n)} A_s = \left\lfloor \frac{s+1}{2} \right\rfloor.$$

In fact the same lower bound holds for recurrences with coefficients in $\mathbb{C}(n)$ that are valid for all sufficiently large n.

Theorem 1.3 (Integral lattice; [source T58])**.** *Let $f = (f_n)_{n\geq 0}$ solve (3). Then*

$$f_n \in \mathbb{Z} \text{ for all } n \geq 0 \quad \Longleftrightarrow \quad f = aS \text{ for some } a \in \mathbb{Z}.$$

Among rational solutions, those for which one integer $D \geq 1$ satisfies $Df_n \in \mathbb{Z}$ for all n are precisely the sequences qS, $q \in \mathbb{Q}$.

These are independent statements. The first does not classify rational solutions of a recurrence, and the second does not prove minimality for all s. Sections 2–5 give the analytic argument. Sections 6–8 give the arithmetic argument, including an explicit prime-adic strengthening. In particular, the lattice theorem can be read and checked without accepting any assertion about analytic continuation.

2 Recurrence width and singularities

We first record the elementary passage from recurrence coefficients to an ordinary differential equation. This passage and the resulting singularity budget are standard tools; the issue in the Franel application is to prove that the candidate singularities are actually present.

Lemma 2.1 (Singularity budget)**.** *Suppose $F(t) = \sum_{n\geq 0} a_n t^n$ has a positive radius of convergence, and a satisfies a recurrence of shift order r over $\mathbb{C}(n)$. There is a nonzero polynomial $H(t)$ of degree at most r such that every nonzero finite point at which a branch continued from the germ of F fails to be holomorphic is a zero of H.*

Proof. Clear denominators and discard finitely many initial indices. We can assume $p_j \in \mathbb{C}[n]$, $p_0 p_r \neq 0$, and that (2) holds for $n \geq N$. Let $\vartheta = t\,d/dt$. Multiplying by t^{n+r} and summing gives

$$LF = R(t), \qquad L = \sum_{j=0}^{r} t^{r-j} p_j(\vartheta - j), \tag{4}$$

where R is a polynomial arising only from the omitted initial indices. Indeed, after putting $k = n + j$, the jth summand becomes $t^{r-j} \sum_{k\geq N+j} p_j(k-j) a_k t^k$.

Let $D = \max_j \deg p_j$ and $c_j = [n^D]p_j(n)$. The coefficient of the highest ordinary derivative d^D/dt^D in L is

$$t^D H(t), \qquad H(t) = \sum_{j=0}^{r} c_j t^{r-j}. \tag{5}$$

At least one c_j is nonzero, and the powers of t in this last sum are distinct. Consequently $H \neq 0$ and $\deg H \leq r$; cancellation cannot destroy the differential operator's highest derivative.

If $t_0 \neq 0$ and $H(t_0) \neq 0$, equation (4) is an ordinary linear differential equation near t_0. Every local solution on a punctured subdomain extends holomorphically through t_0: write the equation as a first-order analytic system and use its analytic fundamental matrix. This applies to every branch obtained by continuation of F, since the differential equation continues with the branch. When $D = 0$, the conclusion follows directly from $F = R/H$. Thus all the specified singularities lie among the roots of H. □

Remark 2.2. The degree D of the recurrence coefficients is unrestricted. It is not the quantity bounded below here. Even a recurrence with extremely large polynomial coefficients has at most r distinct nonzero finite singularities in the sense of Lemma 2.1. Initial exceptions change R, but not this conclusion. Apparent zeros of H are allowed; the lemma gives an upper bound on genuine singularities, not an equality.

3 Definite critical points are visible

The following statement supplies the analytic step used in the application. It explicitly excludes saddle points on the critical fiber and permits a critical value in the interior of the global range.

Definition 3.1. Let M be a compact real-analytic manifold without boundary, of dimension $d \geq 1$, with a positive analytic density $d\mu$. Let $\Phi : M \to \mathbb{R}$ be analytic. A value v is called a *same-side definite value* if the critical points in $\Phi^{-1}(v)$ form a finite nonempty set, every such critical point is nondegenerate, and all their Hessians are positive definite or all are negative definite. Regular points in the same fiber are permitted.

Lemma 3.2 (Local density at a definite value)**.** *Under these hypotheses, near a same-side definite value v the pushforward measure $\nu = \Phi_* \mu$ is absolutely continuous. Its density has the form*

$$\rho(v + t) = r(t) + \mathbf{1}_{\{\epsilon t > 0\}}(\epsilon t)^{d/2-1} h(\epsilon t), \tag{6}$$

where r is C^∞, h is real analytic near zero with $h(0) > 0$, and $\epsilon = 1$ for minima, $\epsilon = -1$ for maxima. In particular ρ does not agree almost everywhere near v with a C^∞ function.

Proof. Choose disjoint coordinate neighborhoods of the finitely many critical points. By the analytic Morse lemma, the function in each neighborhood takes the form

$$\Phi(y) = v + \epsilon(y_1^2 + \cdots + y_d^2),$$

and the density becomes $g(y)\, dy$ with g analytic and $g(0) > 0$. Choose a smooth nonnegative cutoff equal to one on a smaller neighborhood. For small $w > 0$, the entire sphere $|y| = \sqrt{w}$ lies in that smaller neighborhood, so polar coordinates give its exact contribution to the density:

$$\frac{1}{2} w^{d/2-1} \int_{\mathbb{S}^{d-1}} g(\sqrt{w}\, \omega)\, d\omega. \tag{7}$$

The spherical average has a convergent power series in w. All odd-degree Taylor monomials of g integrate to zero, and the even-degree terms give integer powers of w. Its constant term is

positive. Thus each critical point contributes the indicated one-sided power with a positive analytic leading coefficient. All points use the same side ϵ, so those coefficients add.

The complement of the smaller critical neighborhoods is compact. On its part mapping to a sufficiently small interval about v, the gradient is nonzero. Otherwise critical points outside those neighborhoods would accumulate on the fiber $\Phi^{-1}(v)$, contrary to the choice of neighborhoods and compactness. In submersion coordinates, pushforward of a smooth compactly supported density is C^∞: integrate the smooth transverse coordinates, differentiating under the integral. A finite partition of unity therefore gives the function r in (6).

For even d, the exponent $d/2-1$ is an integer $k \geq 0$. The one-sided term has a nonzero jump in its kth derivative. For odd $d \geq 3$, it has a half-integral exponent and fails to be smooth at a finite derivative order. For $d = 1$ it is unbounded like $|t|^{-1/2}$, but is locally integrable. Multiplication by h with $h(0) > 0$ and addition of $r \in C^\infty$ cannot remove any of these defects. This proves the last assertion as well. □

Theorem 3.3 (Visibility criterion; [source T39])**.** *Let*

$$a_n = \int_M \Phi(x)^n \, d\mu(x), \qquad F(t) = \sum_{n \geq 0} a_n t^n.$$

For every nonzero same-side definite value v, the germ of F has a continued branch that fails to extend holomorphically through $t = 1/v$. Consequently any recurrence for a_n over $\mathbb{C}(n)$ has shift order at least the number of distinct nonzero same-side definite values of Φ.

Proof. Since M is compact, Φ is bounded and F converges near zero. Define

$$G(z) = \int_M \frac{d\mu(x)}{z - \Phi(x)} = \int_{\mathbb{R}} \frac{d\nu(u)}{z - u}. \tag{8}$$

The function G is holomorphic in the upper half-plane. For large $|z|$,

$$G(z) = z^{-1} F(z^{-1}), \qquad F(t) = t^{-1} G(t^{-1}).$$

Thus the upper-half-plane branch is connected to the germ at infinity obtained from the original power series. It is not an unrelated solution of a Picard–Fuchs equation.

We show that this branch of G cannot extend holomorphically through v. For $y > 0$,

$$-\frac{1}{\pi} \operatorname{Im} G(x + iy) = \int_{\mathbb{R}} \frac{1}{\pi} \frac{y}{(x-u)^2 + y^2} \, d\nu(u). \tag{9}$$

The right side is the Poisson approximation to the finite measure ν. It converges to ν in distributions as $y \downarrow 0$. For completeness, pair it with a compactly supported smooth test function ψ: the inner integral is its Poisson average and converges uniformly to $\psi(u)$, so dominated convergence against the finite measure proves the claim.

If the indicated branch of G were holomorphic on a full disk centered at v, its imaginary part on the real diameter would be real analytic. The convergence on smaller diameters would then also be smooth, and (9) would force ν there to have the real-analytic density $-\pi^{-1} \operatorname{Im} G(x)$. This contradicts Lemma 3.2. The argument is valid even when ρ is unbounded, because it uses distributional boundary values instead of an unjustified pointwise formula at v itself.

Inversion carries a neighborhood of $v \neq 0$ biholomorphically to a neighborhood of $1/v$. Multiplication by t^{-1} is harmless there. Hence the corresponding branch of F is genuinely singular at $1/v$. Distinct values give distinct points, and Lemma 2.1 proves the recurrence lower bound. □

Remark 3.4 (Why positivity alone would not suffice). A general critical point can be a saddle, and contributions on opposite sides of the same critical value can have different regularity behavior. The proof uses the stronger assertion that *all* critical points on the fiber are definite and have the same side. We will verify this by solving every nonzero critical-point equation of the Franel torus. Checking one attractive critical point while ignoring the others would not prove visibility.

Remark 3.5 (Non-global extrema). The critical point need not maximize or minimize Φ on all of M. The regular points lying over the same value contribute only a smooth density. The continuation path from infinity runs in the upper half-plane; it does not have to pass through intervening real critical values. These two observations are what make interior critical values usable in the singularity count.

4 The Franel torus

Fix $s \geq 2$ and put $d = s - 1$. With $X = x_1 \cdots x_d$, define the Laurent polynomial

$$P_s(x_1, \ldots, x_d) = \prod_{i=1}^{d} (1 + x_i)(1 + X^{-1}). \tag{10}$$

All constant terms below are in the variables $x_1, \ldots, x_d$.

Lemma 4.1 (Exact moment representation). *For every $n \geq 0$, $\mathrm{CT}\, P_s^n = A_s(n)$. On the unit torus $x_i = e^{i\theta_i}$, P_s is the real-valued function*

$$\Phi_s(\theta) = 2^s \prod_{i=1}^{d} \cos(\theta_i/2) \cos((\theta_1 + \cdots + \theta_d)/2). \tag{11}$$

Consequently, with normalized Haar measure on $\mathbb{T}^d$,

$$A_s(n) = \frac{1}{(2\pi)^d} \int_{\mathbb{T}^d} \Phi_s(\theta)^n \, d\theta, \qquad F_s(t) = \frac{1}{(2\pi)^d} \int_{\mathbb{T}^d} \frac{d\theta}{1 - t\Phi_s(\theta)}.$$

The generating-function identity holds for $|t| < 2^{-s}$.

Proof. Expand each factor of P_s^n. If the term chosen from $(1+X^{-1})^n$ has index k, its exponent in every variable is $-k$. A constant term requires index k from each $(1+x_i)^n$. The coefficient is then $\binom{n}{k}^s$, and summing over k proves the first assertion. Constant terms equal torus averages. For the reality assertion use $1 + e^{i\theta} = 2e^{i\theta/2}\cos(\theta/2)$. The phases from the first d factors cancel the phase from the last one, giving (11). Although individual half-angle cosines change sign under a 2π shift, the product is periodic, as is also clear from (10). Finally $|\Phi_s| \leq 2^s$, so the geometric series can be integrated uniformly for the stated values of t. □

Proposition 4.2 (All nonzero critical points). *The critical points of Φ_s with nonzero value are exactly*

$$(x_1, \ldots, x_d) = (\omega, \ldots, \omega), \qquad \omega^s = 1, \quad \omega \neq -1.$$

Their critical value is $(1 + \omega)^s$.

Proof. At a point where $P_s \neq 0$, no factor in (10) vanishes. Logarithmic derivatives, understood locally or simply as the rational derivative P_s'/P_s, give

$$x_i \frac{\partial \log P_s}{\partial x_i} = \frac{x_i}{1 + x_i} - \frac{1}{1 + X}. \tag{12}$$

On the real torus, $\partial/\partial\theta_i = ix_i\partial/\partial x_i$. Since P_s is real there, its real critical equations are precisely the vanishing of (12). Cross-multiplication gives $x_iX = 1$ for each i. Therefore all x_i equal some ω, and $\omega X = \omega^{d+1} = \omega^s = 1$. Conversely each such point satisfies the equations. The nonzero requirement excludes $\omega = -1$. Since $X^{-1} = \omega$ at the point, the value is $(1+\omega)^{d+1} = (1+\omega)^s$. □

Proposition 4.3 (Definiteness and critical values)**.** *Let*

$$m_s = \left\lfloor \frac{s+1}{2} \right\rfloor, \qquad v_j = (-1)^j \big(2\cos(\pi j/s)\big)^s, \quad 0 \le j < s/2.$$

These are exactly the distinct nonzero critical values of Φ_s. Each is a same-side definite value. At $x_i = e^{2\pi ij/s}$, the real Hessian is

$$\operatorname{Hess}_\theta \Phi_s = -(-1)^j \big(2\cos(\pi j/s)\big)^{s-2}(I_d + J_d), \tag{13}$$

where J_d is the all-ones matrix. There are m_s such distinct values.

Proof. At a critical point, direct differentiation of (12) in logarithmic coordinates gives

$$\operatorname{Hess}_{\log x} \log P_s = \frac{\omega}{(1+\omega)^2}(I_d + J_d).$$

Indeed, the diagonal derivative of the first term in (12) is $x_i/(1+x_i)^2$, while differentiating the second term with respect to any $\log x_j$ gives $X/(1+X)^2$. At the critical point the latter equals $\omega/(1+\omega)^2$ as well. Passing to angular variables changes the sign. The gradient is zero, so multiplication by $P_s = (1+\omega)^s$ gives the Hessian of Φ_s, with no additional gradient-square term. For $\omega = e^{2\pi ij/s}$,

$$\frac{\omega}{(1+\omega)^2} = \frac{1}{4\cos^2(\pi j/s)}, \qquad (1+\omega)^s = (-1)^j \big(2\cos(\pi j/s)\big)^s.$$

This proves (13). The eigenvalues of $I_d + J_d$ are 1 with multiplicity $d-1$ and $d+1 = s$ with multiplicity one. The Hessian is therefore nondegenerate and definite. Positive v_j are local maxima and negative v_j are local minima.

Complex conjugation pairs the indices j and $s-j$, without changing the value or Hessian side. The remaining representatives are exactly $0 \le j < s/2$. Their absolute values are strictly decreasing, since cosine is strictly decreasing and positive on $[0, \pi/2)$. Consequently two different representatives cannot give the same value. For odd s there are $(s+1)/2$ representatives; for even s there are $s/2$. Proposition 4.2 excludes any additional critical point on a nonzero fiber, including a saddle. All conditions of the visibility theorem have now been verified. □

5 Proof of the minimum order

Proof of Theorem 1.2. For $s \ge 2$, Lemma 4.1 supplies the real moment representation. Proposition 4.3 provides exactly m_s distinct nonzero same-side definite values. Theorem 3.3 shows that the generating-function germ has genuine continued singularities at all the points $1/v_j$. Lemma 2.1 therefore gives $r \ge m_s$ for every recurrence of shift order r, including those with coefficients in $\mathbb{C}(n)$ and finitely many exceptional initial indices. Stoll's theorem [1] supplies a recurrence over $\mathbb{Q}(n)$ of order at most m_s, proving equality.

For $s = 1$, the binomial theorem gives $A_1(n) = 2^n$ and the order-one relation $A_1(n+1) - 2A_1(n) = 0$. An order-zero relation $p(n)A_1(n) = 0$ with $p \ne 0$ is impossible for all sufficiently large n. This proves the formula also for $s = 1$. □

Example 5.1 (Small parameters)**.** For $s = 2$ the only nonzero critical value is 4, consistent with $A_2(n) = \binom{2n}{n}$ and an order-one recurrence. For $s = 3$ the values are $8, -1$, so two distinct singularities force order at least two. For $s = 4$ the values are $16, -4$. For $s = 5$ there are three values, and the two values apart from 32 are the roots of

$$x^2 + 11x - 1 = 0.$$

Their reciprocals are the roots of $t^2 - 11t - 1 = 0$. This accounts for the nonzero factors $(1-32t)(1+11t-t^2)$ in the highest-degree coefficient polynomial of a corresponding differential equation, up to a nonzero scalar and the convention for writing the shifts. These examples check the signs and inversion in the general proof; they are not its evidence for all s.

Proof-status note 5.2. The crucial additional content beyond an abstract singularity count is Lemma 3.2 and the exhaustive fiber calculation in Propositions 4.2–4.3. No claim that every critical value of every real moment map is singular is used. No irreducibility of a Picard–Fuchs operator is assumed. Nor does minimal order imply that every Apéry auxiliary solution satisfies a particular telescoping recurrence. Those are different statements. External validation of this draft's general minimality argument and its priority remains separate from the proof supplied here.

6 Discrete Abel identities

We now turn to the independent arithmetic argument. A second-order recurrence

$$a_n f_{n+1} = b_n f_n + c_n f_{n-1}, \qquad n \geq 1, \tag{14}$$

with $a_n \neq 0$ is uniquely solved forward from two initial values. For two solutions U, f, define

$$W_n(U, f) = U_n f_{n+1} - U_{n+1} f_n.$$

The following computation is the discrete Abel identity; it is not a new formula. Its use in Apéry settings and the Franel specialization can be found in [4, 3].

Lemma 6.1 (Discrete Abel identity)**.** *Two solutions of* (14) *satisfy*

$$W_n = -\frac{c_n}{a_n} W_{n-1}, \qquad W_n = (-1)^n \left(\prod_{j=1}^{n} \frac{c_j}{a_j} \right) W_0.$$

Proof. Use (14) once for each solution:

$$\begin{aligned} a_n W_n &= U_n(b_n f_n + c_n f_{n-1}) - (b_n U_n + c_n U_{n-1}) f_n \\ &= c_n(U_n f_{n-1} - U_{n-1} f_n) = -c_n W_{n-1}. \end{aligned}$$

Iteration yields the product formula. □

Theorem 6.2 (Prime-denominator rigidity)**.** *Suppose a second-order recurrence has unique forward solutions and an integral solution U with $(U_0, U_1) \neq (0, 0)$. Assume that for every solution f its Casoratian satisfies*

$$W_n(U, f) = \frac{\kappa^n}{(n+1)^m} W_0(U, f), \qquad n \geq 0, \tag{15}$$

where $m \geq 1$ and $\kappa \in \mathbb{Z} \setminus \{0\}$. Then every integral solution is a rational scalar multiple of U. If $\gcd(U_0, U_1) = 1$, it is an integer scalar multiple.

Proof. For integral f, every W_n is an integer. For a prime $p \nmid \kappa$, put $n = p-1$ in (15). The resulting equation

$$p^m W_{p-1} = \kappa^{p-1} W_0$$

forces $p^m \mid W_0$. There are infinitely many such primes. The fixed integer W_0 must therefore vanish. The two initial vectors are proportional over $\mathbb{Q}$; uniqueness propagates that proportionality to the entire sequences. If (U_0, U_1) is primitive, choose integers a, b with $aU_0 + bU_1 = 1$. For $f = qU$, the identity $q = af_0 + bf_1$ then shows $q \in \mathbb{Z}$. □

Remark 6.3. Primitivity matters if the exact answer is written as $\mathbb{Z}U$. Without it, the integral points on the rational line $\mathbb{Q}U$ can include nonintegral multiples of the chosen U. The Franel normalization $S_0 = 1$ makes this issue automatic. Also, two solutions with $W_0 = 0$ are identified through initial-value uniqueness; dividing by terms U_n that might vanish is unnecessary in the general argument.

7 The cubic Franel lattice

The binomial sum $S = A_3$ is integral and has $S_0 = 1, S_1 = 2$. It satisfies the classical recurrence (3), recorded in the original Franel work and in [3, 2]. For any other solution f, Lemma 6.1 gives

$$W_n(S, f) = -\frac{8n^2}{(n+1)^2} W_{n-1}(S, f) = \frac{(-8)^n}{(n+1)^2}(f_1 - 2f_0). \tag{16}$$

The product telescopes because $\prod_{j=1}^{n} j^2/(j+1)^2 = (n+1)^{-2}$.

Proof of Theorem 1.3. Suppose first that f is integral. For each odd prime p, equation (16) at $n = p-1$ implies

$$p^2 \mid f_1 - 2f_0.$$

Thus $f_1 = 2f_0$. The recurrence determines every later term, and the initial vector is $f_0(1, 2)$. Hence $f = f_0 S$, with $f_0 \in \mathbb{Z}$. Conversely every integer multiple of S is an integral solution.

For the second assertion, if Df is integral, it is also a solution. The first assertion gives $Df = aS$ for $a \in \mathbb{Z}$, so $f = (a/D)S$. Conversely a fixed denominator for a rational scalar clears every term of qS. □

The recurrence is imposed for $n \geq 1$, exactly as in the question. Were it imposed at $n = 0$ as well, it would immediately force $f_1 = 2f_0$, making the question trivial. We do not add that condition. Instead, integrality over all indices forces precisely that missing initial relation.

Corollary 7.1 (No other integral ray)**.** *Let $\mathcal{V} \subset \mathbb{Q}^{\mathbb{N}_0}$ be the solution space of (3). Then*

$$\dim_{\mathbb{Q}} \mathcal{V} = 2, \qquad \mathcal{V} \cap \mathbb{Z}^{\mathbb{N}_0} = \mathbb{Z}S,$$

and the bounded-denominator subspace is $\mathbb{Q}S$.

Proof. The recurrence has nonzero forward coefficient $(n+1)^2$, so the initial-value map $\mathcal{V} \to \mathbb{Q}^2$, $f \mapsto (f_0, f_1)$, is an isomorphism. The lattice statements are Theorem 1.3. □

8 Prime-adic strengthening and denominator certificates

The use of infinitely many primes can be replaced by arbitrarily large powers of one fixed prime. This gives information that a mere common-denominator classification does not display.

Proposition 8.1 (Every odd prime detects a second direction)**.** *Let f be a rational solution of (3), and put $c = f_1 - 2f_0$. If $c \neq 0$, then for every odd prime p and every $a \geq 1$,*

$$\min\{v_p(f_{p^a-1}), v_p(f_{p^a})\} \leq v_p(c) - 2a, \tag{17}$$

where $v_p(0) = +\infty$. In particular, the denominators of f_n have unbounded powers of every odd prime.

Proof. At $n = p^a - 1$, equation (16) gives

$$v_p(W_{p^a-1}) = v_p(c) - 2a,$$

since $p \nmid 8$. Both S_{p^a-1} and S_{p^a} are integers. The nonarchimedean triangle inequality therefore gives

$$\begin{aligned} v_p(W_{p^a-1}) &= v_p(S_{p^a-1}f_{p^a} - S_{p^a}f_{p^a-1}) \\ &\geq \min\{v_p(f_{p^a}), v_p(f_{p^a-1})\}. \end{aligned}$$

Rearranging proves (17). Its right side tends to $-\infty$. For a rational number written in lowest terms, a negative valuation $-b$ means its denominator is divisible by p^b, proving the final assertion. □

Corollary 8.2 (One-prime boundedness criterion)**.** *For a rational solution f, the following are equivalent:*

(i) $f \in \mathbb{Q}S$;

(ii) the sequence (f_n) is bounded in $\mathbb{Q}_p$ for some odd prime p;

(iii) it is bounded in $\mathbb{Q}_p$ for every odd prime p.

Also, if $\mathcal{P}$ is a finite set of primes, the solutions with all terms in $\mathbb{Z}[\mathcal{P}^{-1}]$ are exactly $\mathbb{Z}[\mathcal{P}^{-1}]S$.

Proof. For $f = qS$, integrality of S gives $v_p(f_n) \geq v_p(q)$ uniformly. Conversely Proposition 8.1 excludes boundedness at even one odd prime when $c \neq 0$. For the localization statement, choose an odd prime outside $\mathcal{P}$. Every element of the localization is integral at that prime, so the one-prime criterion gives $f = qS$. The equality $q = f_0$ puts q in the same localization. The converse is immediate. □

Example 8.3 (A concrete second solution)**.** Let B solve (3) with $B_0 = 0$, $B_1 = 1$. Exact recurrence gives

$$B_2 = 4, \quad B_3 = \frac{208}{9}, \quad B_4 = \frac{1280}{9}.$$

Here $c = 1$. The denominator 3^2 is already visible at $n = 3$. In general, for every odd prime p, at least one of B_{p-1}, B_p has denominator divisible by p^2, and the pair at $p^a - 1, p^a$ contains a denominator divisible by p^{2a}. Thus early integral values such as B_1, B_2 give no evidence for integrality of the full sequence.

Remark 8.4 (A finite certificate from initial values). For a nonzero integer $c = f_1 - 2f_0$, any odd prime p with $p^2 \nmid c$ certifies that the entries cannot all be integral through index p. Indeed $W_{p-1} = (-8)^{p-1}c/p^2$ is then nonintegral while its defining determinant would be integral if all those entries were integral. This is a direct, checkable certificate; it does not require the complete factorization of a large later denominator.

9 Scope and provenance of the two arguments

The source audit identifiers serve only to connect this paper with its preparatory notes. They are not external theorem numbers.

Source	Result retained here	Essential justification
T39	Definite-value visibility	Positive same-side Morse density and Poisson boundary values; every critical point in the fiber is covered.
T40	General minimal shift order	Exhaustive torus critical-point and Hessian calculation, visibility, singularity budget, Stoll upper bound.
T58	Integral solution lattice	Discrete Abel identity and divisibility of the one initial determinant by arbitrary odd-prime powers.

The singularity budget is a standard recurrence-to-differential-equation argument. The Morse lemma, the positivity of pushforward measures and Poisson boundary convergence are classical analysis. The proposed contribution in the first part is their complete specialization to the Franel torus, including all nonzero critical fibers. The discrete Abel formula in the second part is also classical. Its elementary prime-denominator consequence is what produces the full integral-lattice classification and the stronger prime-adic statement.

The two outputs measure different kinds of rigidity. The minimum order says that a sequence cannot be described by a shorter rational-coefficient relation. The lattice statement says that one fixed two-dimensional rational recurrence space has only one integral direction. Neither assertion is a transcendence theorem for a named constant. In particular, no conclusion about irrationality of an arbitrary Apéry limit follows merely from unbounded denominators of a second solution: such a conclusion also requires quantitative approximation estimates of the appropriate strength.

No exhaustive priority certificate is asserted. The original general lower bound is explicitly separated from the telescoping version in [2]; Tonien records the remaining integral-sequence question in [3]. The proofs above can therefore be evaluated against clearly specified existing questions. Independent review must assess the argument and its place in the literature rather than rely on the historical labels attached to the source notes.

A A compact proof audit

This appendix identifies the exact points at which a tempting shorter argument could fail, and the corresponding justification supplied in the paper.

1. **Reality of the moment map.** The complex phases in (10) cancel on the full torus. A generic Laurent polynomial would not produce the real positive-measure Cauchy transform needed in Theorem 3.3.

2. **Completeness of the critical list.** The derivation divides by P_s only on a nonzero fiber. It proves that every point on such a fiber is either regular or one of the listed diagonal roots of unity. Degenerate critical points at value zero do not enter the finite nonzero singularity budget.

3. **No hidden saddle contribution.** Equation (13) and the exhaustive list imply definiteness of all critical points on the fiber. Conjugate points contribute the same side and

a positive leading density.

4. **Smoothness of the remaining density.** Compactness supplies a whole interval on which the cutoff complement is a submersion. The proof requires only a C^∞ remainder, not an analytic partition of unity.

5. **A singularity of the correct branch.** The upper half-plane connects the Cauchy transform at infinity to every real critical value from above. Distributional Poisson convergence rules out a holomorphic extension there. A local singularity of an unrelated period would be insufficient.

6. **Width rather than differential order.** The top differential coefficient is $t^D H(t)$ with $\deg H \le r$, even when D is much larger than r. The number of nonzero singular points bounds r.

7. **Initial indices of the lattice question.** The recurrence starts at $n = 1$. The relation $f_1 = 2f_0$ is deduced from integrality and is not imposed as an additional initial constraint.

8. **Fixed versus varying denominators.** Rationality of each term alone is automatic and gives a two-dimensional space. A common denominator is a genuinely stronger requirement. Proposition 8.1 quantifies how it fails away from the Franel line.

References

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